

Why AI Models Sometimes Give Wrong or Made-Up Information

Artificial intelligence can answer questions within seconds, explain complicated ideas, summarize documents, and generate detailed responses. However, an answer that sounds clear and confident is not necessarily an accurate one. Large language models (LLMs) can sometimes produce information that is false, unsupported, or inconsistent while presenting it as if it were correct. This problem is commonly described as an AI “hallucination.” The National Institute of Standards and Technology (NIST) uses the term “confabulation” for cases in which generative AI produces and confidently presents erroneous or false content. Understanding why these errors occur is important as AI systems become increasingly common in information-seeking tasks.

What Is an AI Hallucination? An AI hallucination occurs when a model generates content that does not accurately correspond to facts, evidence, or the information provided in the prompt. NIST describes confabulation as the generation of confidently stated but erroneous or false content. It also notes that these outputs can contradict information given in the prompt or even contradict statements generated earlier in the same response.

The problem does not mean that an AI system is intentionally trying to deceive the user. It is connected to the way generative models produce language. NIST explains that generative models approximate statistical patterns in their training data. In the case of language models, this includes predicting the next token or word in a sequence. This process can produce accurate and consistent information, but it can also produce factually inaccurate or internally inconsistent information.

Researchers also distinguish between different forms of hallucination. A 2025 survey published in the ACM Transactions on Information Systems divides hallucinations broadly into factuality hallucination and faithfulness hallucination. Factuality hallucination concerns disagreement between generated content and verifiable real-world facts, while faithfulness hallucination concerns inconsistency with the user's instructions, supplied context, or the model's own previous statements.

Why Do AI Models Make These Errors? One important reason is that language models learn statistical patterns rather than functioning as perfectly organized databases of verified facts. During training, models learn relationships between words, phrases, concepts, and other patterns in large collections of data. When responding to a prompt, they use what they have learned to generate a likely sequence of language.

This approach is extremely useful for producing natural language, but it does not guarantee factual accuracy. NIST explains that statistical prediction can produce both accurate and inaccurate outputs. The problem can become especially important for open-ended questions and areas requiring highly specific or contextual knowledge.

A 2026 study published in Nature provides another explanation. The researchers argue that next-word prediction creates statistical pressure toward hallucination. Their analysis suggests that recurring patterns, such as grammar, are easier for models to learn reliably, whereas rare or one-off facts can be more difficult to reproduce accurately. The researchers argue that this tendency can exist even when the training data itself is assumed to be error-free.

However, this does not mean that every hallucination has one single cause. A 2024 survey on LLM factuality identifies multiple challenges and possible causes and examines problems connected with data, model training, generation, and factuality evaluation. A separate 2024 empirical study also examines hallucinations across different stages of model development and use, including their sources, detection, and mitigation.

Why Do Wrong Answers Sound Convincing? One of the most difficult aspects of AI hallucination is that an incorrect answer can still sound professional, detailed, and confident. Fluency and factual accuracy are different properties.

An AI model can produce grammatically correct sentences and a logical-looking explanation without having verified every claim in that explanation. NIST specifically warns that the confident nature of some generated content can cause people to trust false information. It also notes that models can sometimes generate false citations or reasoning that appear to support an incorrect answer.

This creates an important distinction between language generation and factual verification. A response may be linguistically convincing without being factually reliable. Therefore, the confidence or quality of the writing itself cannot be treated as proof that the information is correct.

How Are Hallucinations Measured? Researchers have developed different methods to measure factual accuracy instead of judging an AI answer only by how convincing it sounds.

One important example is Google DeepMind's research on long-form factuality in large language models. The researchers created LongFact, a collection of thousands of fact-seeking questions covering 38 topics. They also developed SAFE (Search-Augmented Factuality Evaluator), which breaks a long response into individual factual claims and uses search to examine whether those claims are supported by evidence.

The researchers evaluated 13 language models from four model families. Their results showed that larger language models generally achieved better long-form factuality on their evaluation. However, this does not mean that larger models never make factual errors. It only describes the results of the particular evaluation used in the study.

Other researchers have developed additional benchmarks. The 2024 HaluEval 2.0 study investigated hallucination detection and mitigation and examined potential factors contributing to hallucinations. More recent work such as FactBench uses an evidence-based evaluation pipeline called VERIFY to classify generated content according to whether it is supported, unsupported, or undecidable based on web evidence. The study also reports that factual precision decreases as prompts become more difficult and that greater model scale does not necessarily guarantee better factuality in every comparison.

Does a Larger AI Model Always Make Fewer Errors? The research does not support a simple answer that larger models always make fewer errors.

The DeepMind LongFact study found that larger models generally performed better in its long-form factuality evaluation. However, FactBench reported that factuality does not necessarily improve with scale in every model comparison. For example, the researchers found a case in which a larger Llama model performed comparably to or worse than its smaller counterpart on their benchmark.

These findings are not necessarily contradictory because the studies use different models, benchmarks, prompts, and evaluation methods. They demonstrate why model size alone should not be treated as a complete measure of factual reliability.

Therefore, the question is more complicated than simply asking whether an AI model is “bigger.” The quality of training, the type of task, the information being requested, the evaluation method, and the model's ability to verify information can all matter.

Why Do Models Sometimes Guess Instead of Admitting Uncertainty? A particularly interesting issue concerns how AI systems are evaluated.

A 2026 Nature study argues that accuracy-based evaluations can unintentionally encourage models to guess. In many traditional evaluations, an answer is marked either correct or incorrect, while refusing to answer or expressing uncertainty receives no credit. Under such a system, guessing can sometimes be more rewarding than admitting uncertainty.

The researchers propose open-rubric evaluations, in which the scoring rules are explicitly stated and the consequences of incorrect answers or abstaining are made clear. Their argument is that evaluation should allow models to adjust their willingness to answer according to the consequences of being wrong.

This provides a different perspective on hallucination. The problem may not only concern what information a model has learned. According to this research, the way a model is evaluated can also influence whether it tends to guess or express uncertainty.

At the same time, this is a research proposal rather than a universal solution. The authors themselves discuss limitations of open-rubric evaluation, including the fact that real users do not normally provide an explicit confidence threshold with every question.

Can Hallucinations Be Reduced? Researchers have investigated several approaches for reducing factual errors. One important approach is retrieval-augmented generation (RAG).

RAG combines a language model with an information-retrieval system. Instead of relying only on information encoded in the model's parameters, the system retrieves relevant external information and uses it during response generation.

The original RAG research by Lewis and colleagues, published at NeurIPS 2020, combined a pre-trained sequence-to-sequence model with a non-parametric memory containing retrieved information from Wikipedia. On the knowledge-intensive language-generation tasks they evaluated, the RAG models produced more specific, diverse, and factual language than the parametric-only baseline.

This demonstrates why external evidence can be useful. If a model can retrieve relevant information, it does not have to depend entirely on what was learned during its original training.

However, RAG is not a perfect solution. The 2025 ACM survey specifically discusses limitations of retrieval-augmented systems and emphasizes that hallucinations can still occur in RAG-based systems.

Other approaches include fact-checking, uncertainty estimation, consistency-based verification, better training methods, and improved evaluation. Research on hallucination mitigation continues to examine which approaches work best under different circumstances.

What Does Current Research Agree and Disagree About? There is broad agreement across the research that hallucination is a significant factuality problem for LLMs. NIST identifies confabulation as a risk of generative AI, while academic research has developed multiple benchmarks and methods for studying the problem.

There is also evidence that several approaches can improve factuality. Retrieval can provide external information, factuality benchmarks can identify unsupported claims, and evaluation methods can be designed to account for uncertainty.

However, researchers do not provide one universally accepted explanation or solution. For example, the DeepMind study found that larger models generally performed better on its factuality benchmark, while FactBench found that factuality did not necessarily improve with scale in every comparison.

Similarly, RAG can improve factual generation in certain tasks, but research reviews continue to identify limitations in retrieval-augmented systems.

The 2026 Nature study adds another dimension by arguing that evaluation itself can encourage guessing. This suggests that improving AI reliability may require changes not only to models and training but also to the way systems are tested and rewarded.

Methodology Note For this investigation, sources were selected from government guidance, peer-reviewed academic research, conference proceedings, and established research organizations. The sources were chosen because they directly address the causes, measurement, evaluation, or mitigation of hallucinations in large language models.

Primary research was given particular attention. These sources include the DeepMind LongFact study, the original RAG research, the HaluEval 2.0 study, the FactBench study, the FactSelfCheck study, and the 2026 Nature research on evaluation incentives.

The findings were compared rather than treated as identical. Where studies produced different results—for example, regarding the relationship between model size and factuality—the differences were reported with attention to their different benchmarks and experimental settings. Claims were not extended beyond what the individual sources evaluated.

Conclusion AI models sometimes give wrong or made-up information because generating fluent language and verifying facts are different tasks. Large language models learn statistical patterns and generate likely sequences of language, but this process does not guarantee that every generated claim is true.

Research identifies several factors connected to hallucination, including difficulties with rare facts, limitations related to data and model development, and the challenges of evaluating factuality. Research also shows that the way AI systems are evaluated can influence whether they guess or express uncertainty.

Several methods can improve factual reliability. LongFact and other benchmarks provide ways to measure factual claims, while retrieval-augmented generation gives models access to external information. However, no single method completely eliminates hallucinations.

The evidence therefore suggests that AI reliability is a continuing research problem rather than a problem with one simple solution. Larger models can perform better on some factuality evaluations, retrieval can provide useful external evidence, and uncertainty-aware evaluation can discourage inappropriate guessing. At the same time, errors remain possible. For users, this means that a fluent or confident AI response should not automatically be treated as verified information.

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